

R Code

```
1 # Daily COVID counts, Our World in Data.
2
3 url <- paste0(
4   "https://raw.githubusercontent.com/owid/covid-19-data/",
5   "master/public/data/owid-covid-data.csv"
6 )
7
8 download.file(url, "owid-covid-data.csv")
9
10 owidall <- read.csv("owid-covid-data.csv")
11
12 # Check that the data were loaded
13 head(owidall)
14
15 # Remove aggregate records and keep country-level data
16 owidall <- owidall[!grepl("^OWID", owidall$iso_code), ]
17
18 # Subset by continent
19 owideu <- subset(owidall, continent == "Europe")
20 head(owideu)
21
22 # Check number of rows and columns
23 dim(owideu)
24
25 # Check dataset is of Europe
26 unique(owideu$continent)
27
28 # Check the names of all variables
29 names(owideu)
30
31 # Check the list of countries
32 unique(owideu$location)
33
34 # Check the date range
35 range(as.Date(owideu$date))
36
37 # Sort observations from the highest to lowest and display country, date, and number
  of deaths
38
39 owideu[
40   order(-owideu$new_deaths),
41   c("location", "date", "new_deaths")
42 ]
43
44 # Convert date format
45 owideu$date <- as.Date(owideu$date)
46
47 # Set up plot area, suppress axes
48 plot(
49   owideu$date,
```

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50     owideu$new_deaths,
51     type = "p",
52     pch = 16,
53     cex = 0.5,
54     col = "deeppink",
55     xaxt = "n",
56     yaxt = "n",
57     xlab = "",
58     ylab = "COVID Deaths in Europe (Daily)",
59     ylim = c(0, max(owideu$new_deaths, na.rm = TRUE) * 1.15)
60 )
61
62 # Y-axis ticks
63 y_ticks <- seq(
64     0,
65     max(owideu$new_deaths, na.rm = TRUE),
66     by = 1000
67 )
68
69 axis(
70     2,
71     at = y_ticks,
72     las = 1
73 )
74
75 # X-axis ticks
76 x_ticks <- seq(
77     min(owideu$date),
78     max(owideu$date),
79     by = "1 month"
80 )
81
82 axis(
83     1,
84     at = x_ticks,
85     labels = format(x_ticks, "%Y-%m"),
86     las = 2
87 )
88
89 # X-axis title
90 mtext(
91     "Date",
92     side = 1,
93     line = 4
94 )
95
96 # Find the highest daily death counts
97 sorted <- owideu[
98     order(-owideu$new_deaths),
99     c("location", "date", "new_deaths")
100 ]
101
102 peaks <- data.frame()
103 remaining <- sorted

```

```
104
105 while (nrow(remaining) > 0 && nrow(peaks) < 5) {
106   top <- remaining[1, ]
107   peaks <- rbind(peaks, top)
108
109   # Remove nearby observations from the same country
110   remaining <- remaining[
111     !(remaining$location == top$location &
112       abs(as.numeric(remaining$date - top$date)) < 60),
113     ]
114 }
115
116 # Add country labels at the peak points
117 text(
118   peaks$date,
119   peaks$new_deaths,
120   labels = peaks$location,
121   pos = 3,
122   cex = 0.7
123 )
```